Worksheet

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Measurement

We want to know whether people in Toronto are sleeping worse than they were ten years ago. Write down three different ways that "sleeping worse" could be measured. For each, say what the instrument is, what the units are, and give one reason to doubt its validity and one reason to doubt its reliability.

The City of Toronto's daily shelter occupancy data, which we used in Part 1, record the number of people in each shelter program each night. Someone uses them as a measure of homelessness in Toronto. Who is counted that should not be, who is not counted that should be, and in which direction does each push the number?

For each of the following say whether the data are censored, truncated, or missing, and what would go wrong if we ignored it.

a. A survey records income in bands, and the top band is "$200,000 or more".
b. A study of hospital stays uses records of patients who were discharged in 2025.
c. A bathroom scale that reads up to 150 kg.
d. A question about drug use that one in five respondents skip.

Population, frame, and sample

For one of the surveys below, write down the target population, the sampling frame, and how the sample is selected. Then write down one group that the survey leaves out of its target population by definition, one group that is in the target population but is likely to be missed by the frame, and one way that those who respond might differ from those who were selected. You will need to find and read the methodology documentation.

1. The Canadian Labour Force Survey, which produces the monthly unemployment rate.
2. The US General Social Survey.
3. A poll about the Toronto mayoral election conducted on an online opt-in panel, reported in a newspaper this month.

Bowley in Reading

These questions are about Bowley (1913).

How did Bowley choose which households to visit? Which of the designs from the lecture is it closest to, and what would have had to be true of his list for that to cause a problem?

Bowley had to decide what a "working-class household" was before he could sample one. Find the definition in the paper. Name one kind of household that you think his definition would have missed, and say how that would affect his estimate of poverty.

What does Bowley say about how far wrong his estimates could be because they come from a sample? Put his argument in your own words and write two or three sentences.

What could Bowley claim about Reading? What could he claim about England? What is the difference?

Neyman

These questions are about Sections I, III, and V of Neyman (1934).

What is purposive selection, and what is the case for it? Make the best argument for it that you can in three sentences.

What went wrong with Gini and Galvani's sample of the Italian census, according to Neyman?

A polling firm recruits respondents online, and then weights them so that the sample matches the census on age, gender, region, and education. Is this closer to random or to purposive selection? What would Neyman want to know before trusting it?

References

Bowley, Arthur Lyon. 1913. "Working-Class Households in Reading." *Journal of the Royal Statistical Society* 76 (7): 672–701. https://doi.org/10.2307/2339708.

Neyman, Jerzy. 1934. "On the Two Different Aspects of the Representative Method: The Method of Stratified Sampling and the Method of Purposive Selection." *Journal of the Royal Statistical Society* 97 (4): 558–625. https://doi.org/10.2307/2342192.